# Practical No. 4

**Aim:** To study and execute the DCL command.

**Software Required:** MySQL, Oracle 10g

**Theory:**

**DCL Full Form – Data Control Language**

**DCL commands** are used to control privileges in the database. The privileges (right to access the data) are required for performing all the database operations, like creating tables, views, or sequences. DCL statements are used to perform the work related to the **rights, permissions**, and **other control** of the database system.

Commands granted to the user as a Privilege List are: UPDATE, SELECT, ALTER, DELETE.

Commands in DCL are: **GRANT** and **REVOKE.**

1. **GRANT**

   This command is used to grant permission to the user to perform a particular operation on a particular object. If we are a **database administrator** and we want to restrict user accessibility such as one who only views the data or may only update the data, we can give the privilege permission to the users according to our wish.

   **Syntax:** GRANT privilege_list

   ON Object_name

   TO user_name;

   **Example:** GRANT SELECT, INSERT ON college.students TO 'root'@'localhost';

2. **REVOKE**

   This command is used to take permission/access back from the user. If we want to return permission from the database that we have granted to the users at that time we need to run REVOKE command.

**Syntax:** REVOKE privilege_list

ON object_name

FROM user_name;

**Example:** REVOKE INSERT ON college.students FROM 'root'@'localhost';

**Program&Output:**

**For ADMIN user:**

CREATE DATABASE class;

USE class;

CREATE TABLE student(ID int, Name varchar(50), Brach varchar(50));

INSERT INTO student VALUES (1, 'Iron Man', 'CSE');

INSERT INTO student VALUES (2, 'Spiderman', 'AI');

INSERT INTO student VALUES (3, 'Hulk', 'CE');

SELECT * FROM student;

CREATE USER 'tony'@'localhost' IDENTIFIED BY 'tony123';

GRANT USAGE ON *.* TO ' tony'@'localhost';

GRANT SELECT, INSERT ON class.student TO ' tony'@'localhost';

SHOW GRANTS FOR 'tony'@'localhost';

REVOKE INSERT ON class.student FROM ' tony'@'localhost';

REVOKE SELECT ON class.student FROM ' tony'@'localhost';

```
mysql> CREATE DATABASE class;
Query OK, 1 row affected (0.14 sec)

mysql> USE class;
Database changed
mysql> CREATE TABLE student(ID int, Name varchar(50), Brach varchar(50));
Query OK, 0 rows affected (0.49 sec)

mysql> INSERT INTO student VALUES (1, 'Iron Man', 'CSE');
Query OK, 1 row affected (0.06 sec)

mysql> INSERT INTO student VALUES (2, 'Spiderman', 'AI');
Query OK, 1 row affected (0.08 sec)

mysql> INSERT INTO student VALUES (3, 'Hulk', 'CE');
Query OK, 1 row affected (0.08 sec)

mysql> SELECT * FROM student;
+------+-----------+-------+
| ID   | Name      | Brach |
+------+-----------+-------+
|    1 | Iron Man  | CSE   |
|    2 | Spiderman | AI    |
|    3 | Hulk      | CE    |
+------+-----------+-------+
3 rows in set (0.00 sec)

mysql> CREATE USER 'tony'@'localhost' IDENTIFIED BY 'tony123';
Query OK, 0 rows affected (0.13 sec)

mysql> GRANT USAGE ON *.* TO ' tony'@'localhost';
Query OK, 0 rows affected (0.13 sec)

mysql> GRANT SELECT, INSERT ON class.student TO ' tony'@'localhost';
Query OK, 0 rows affected (0.19 sec)

mysql> SHOW GRANTS FOR 'tony'@'localhost';
+-----------------------------------------------------------------+
| Grants for tony@localhost                                       |
+-----------------------------------------------------------------+
| GRANT USAGE ON *.* TO `tony`@`localhost`                        |
| GRANT SELECT, INSERT ON `class`.`student` TO `tony`@`localhost` |
+-----------------------------------------------------------------+
2 rows in set (0.00 sec)

mysql> REVOKE INSERT ON class.student FROM ' tony'@'localhost';
Query OK, 0 rows affected (0.09 sec)

mysql> REVOKE SELECT ON class.student FROM ' tony'@'localhost';
Query OK, 0 rows affected (0.07 sec)
```

**For NEW user:**

USE class;

SELECT * FROM student;

INSERT INTO student VALUES(4, 'Dr. Strange' , 'DS' );

DELETE FROM student WHERE ID = 1;

INSERT INTO student VALUES(5, 'Antman' , 'IT' );

SELECT * FROM student;

```
mysql> USE class;
Database changed
mysql> SELECT * FROM student;
+------+----------+-------+
| ID   | Name     | Brach |
+------+----------+-------+
|    1 | Iron Man | CSE   |
|    2 | Spiderman | AI    |
|    3 | Hulk     | CE    |
+------+----------+-------+
3 rows in set (0.00 sec)

mysql> insert into student values(4, 'Thanos', 'CE');
Query OK, 1 row affected (0.10 sec)

mysql> DELETE FROM student WHERE ID = 1;
ERROR 1142 (42000): DELETE command denied to user 'tony'@'localhost' for table 'student'
mysql> INSERT INTO student VALUES(5, 'Antman', 'IT');
ERROR 1142 (42000): INSERT command denied to user 'tony'@'localhost' for table 'student'
mysql> SELECT * FROM student;
ERROR 1142 (42000): SELECT command denied to user 'tony'@'localhost' for table 'student'
mysql>
```

**Conclusion:** Hence, we have successfully studied and implemented DCL commands.